Interpretive Cooperation in the Age of Algorithmic Semiosis

Postdoctoral project

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Summary

Generative systems deliver objects that are *already interpreted*: the inferential walk that once belonged to the reader is pre-executed by computation. This project describes that regime — **algorithmic semiosis** — through three gestures, and draws out its consequences for the university as a site of interpretation.

Mapping, dwelling, resisting: toward a semiotic reform of higher education.

Mapping

Describe

Turning probabilistic reliefs into a legible map.

- statistical attractor
- perceptual attractor

- probabilistic bifurcation
- the thinking-visible
- real perceptual operators

Dwelling

Protect

Holding the not-yet-resolved against premature closure.

- semiotic silence
- *Ma*
- inhabitation
- hesitation
- transitional space (Winnicott)

Resisting

Act

Restoring to the interpreter the position of one who searches.

- semiotic resistance
- the second inquiry
- the inferential walk, resumed
- emancipation (Rancière)
- a pedagogy of indeterminacy

The conceptual vocabulary above is developed at length in the *Glossaire de la sémiose algorithmique*, currently available in French only.

The problem

My doctoral thesis (*L'accessibilité esthétique des mondes possibles au cinéma*, UQAM, 2022 (Prud'homme 2022)) developed the concept of *accessibility operators*: the double-addressed semiotic signals by which characters and viewers cross the boundaries between fictional worlds. This project examines the contemporary mutation of that mechanism. In the age of generative AI, the devices that orient semiosis spill beyond fictional universes and act as *real perceptual operators*: generative images, LLM outputs, automated syntheses and pedagogical interfaces reconfigure the conditions of semiosis itself. The boundary has shifted, and now runs between the interpreting subject and the semiotic object.

The pivotal concept is Eco's *interpretive cooperation* (Eco [1979] 1985): every semiotic object is a lazy machine whose gaps call for the reader's inferential journey. Generative AI performs the inverse operation — what I call **reversed interpretive cooperation**. Optimised to converge on an audience's semiotic attractors, generative systems deliver an object that arrives already interpreted, its credibility immanent to its form. The inferential walk that was the interpreter's work becomes an operation pre-executed by computation; the journey is replaced by a pre-calculated destination. Semiotics describes this mechanism with a precision that neither philosophy of technology nor cognitive science reaches alone: Peirce's unlimited semiosis (Peirce 2009) gives way to probabilistic convergence, where the index becomes icon by computation and Greimasian veridiction (Greimas and Courtés 1979) is pre-installed in the form. Where Deleuze (Deleuze 1985) conceived the image-thought as a putting-into-crisis of thought, the algorithmic regime anticipates the crisis in order to defuse it: indeterminacy becomes an anomaly to be corrected.

The stakes are institutional. The university has cultivated interpretive cooperation since its origins: reading difficult texts, tolerating indeterminacy (Eco 1965), defending and revising one's inferences. When AI short-circuits that journey, it raises far more than a question of academic integrity — it forces a rethinking of the very purposes of education, that is, of the kind of semiotic subject the university sets out to form.

Three gestures

The project describes *algorithmic semiosis* through three complementary gestures.

Mapping

The diagnostic gesture: describing the generative milieu as a territory of probabilistic reliefs — attractors, overweightings, bifurcations — in which a prompt converges on pre-stabilised forms. The *thinking-visible*, the regime of generative images in which form resolves its own interpretant, and *algorithmic textuality*, its discursive counterpart, appear here as reliefs on the map.

The properly semiotic contribution is to describe the chain of transduction running from the statistical attractor to the perceptual attractor: the passage from index to iconic sign, mediated by computation. The emergence of sparse autoencoders (SAEs) (Cunningham et al. 2023; Templeton et al. 2024), theorised here as partial renderings of Eco's computational encyclopaedia (Eco 1992), provides novel empirical purchase on this gesture. That purchase calls for caution, and the project treats it as an object of analysis as much as an instrument: these devices are optimised for reconstruction, an objective that leaves open whether the features they yield correspond to humanly valid concepts. The veridictory status of this legibility remains to be established, which makes it a semiotic object in its own right.

Dwelling

The anthropological gesture: naming what the proliferation of algorithmic agents makes visible by contrast. *Semiotic silence* designates the interval between the perception of a sign and its resolution into meaning — a structural condition of interpretive creativity, grounded in phenomenological boredom (Husserl, Heidegger), the Japanese *Ma* (Saito 1997), the Lotmanian boundary (Lotman 2000) and Eco's hesitation (Eco 1992). Its inverse is *premature closure*, the structural pathology of agentic architectures in which each cycle calls forth the next.

From this gesture comes the hinge concept of *inhabitation* — dwelling in a semiotic space without filling it — which connects the anthropology of interpretation to the institutional question: the university functions as a transitional space in Winnicott's sense (Winnicott 1971), and it is precisely this in-between that the algorithmic semiosphere threatens to saturate.

Resisting

The political-pedagogical gesture, and the project's most original contribution. *Semiotic resistance* is a critical stance: Rancière's emancipation of the gaze (Rancière 2000), extended to the interpreting subject as a whole. Its operative protocol is the *second inquiry* (*enquête seconde*): an "algorithmic *Blow Up*" that turns generated output from destination into probe and resumes the pre-executed inferential walk. The generative system functions as an explicator in the sense of *The Ignorant Schoolmaster* (Rancière 1987); the second inquiry is the counter-conduct that restores to the interpreter the position of one who searches.

The third gesture returns to the first through an asymmetry that distinguishes the spiral from a circle. Strategic mapping belongs to the algorithmic apparatus, which sets the attractors and imposes its reliefs as self-evident; tactical mapping belongs to the subject who rereads the map as a set of locatable choices. To resist is to counter-map, in the sense in which De Certeau calls tactical the reappropriation, from a position without a proper place, of an apparatus devised elsewhere (Certeau et al. 1980).

Deliverables: a diagnosis of the state of higher education in the face of AI; a set of devices for a pedagogy of indeterminacy that treats hesitation, silence and slowness as competences rather than defects; and a contribution to curriculum reform grounded in the formation of a critical semiotic subject.

Hinge — the interpretive spiral

The three gestures coil rather than follow one another: mapping makes dwelling and resisting possible, which in turn produces new reliefs to map. This open loop is the structural opposite of premature closure — the project's methodological principle as much as its theoretical wager.

Method

Empirical inquiry runs across all three gestures as a transversal method rather than a separate strand. It combines three approaches: semiotic analysis of forms, describing the dominant attractors of a generated corpus; analysis of devices, treating interfaces as real perceptual operators; and interviews with students and instructors, documenting the situated experience of the inferential short circuit.

Corpus and fieldwork

Four types of paradigmatic object:

Generative texts and images from current-generation models, interrogated for their dominant attractors — which forms are overweighted, which bodies, faces and spaces are produced as "natural" or "universal".

University cognitive interfaces — platforms, automated marking, LLM assistants — analysed as real perceptual operators that reconfigure the act of learning upstream of any particular use.

AI-assisted student work, read for traces of the inferential short circuit.

Documented agentic incidents — public code repository archives, issue threads, publications produced by agents, with the OpenClaw / Matplotlib incident of February 2026 as the entry case. This family makes algorithmic semiosis observable in its traces rather than in its finished products alone. Interactions conducted with generative systems in the course of the research are added to the corpus and documented as cases.

Contribution and stance

Dominant discourse on AI and education oscillates between integrative enthusiasm and regulatory panic, two positions that treat AI as a neutral tool whose use must be managed: the managerial register this project takes as its object of critique. The project holds that generative AI acts as a real perceptual operator, reconfiguring the conditions of seeing, reading, thinking and believing independently of use. The pedagogical question then becomes: what semiotic ecology must the university build so that human beings remain the subjects of their own semiosis?

Situating the project within critical studies of digital education

The field of *critical digital education studies* has established a robust analysis of government by measurement: Ben Williamson's work on the datafication of education systems (Williamson 2017), Siân Bayne's and Jen Ross's on digital pedagogies (Bayne et al. 2020; Ross 2023), Neil Selwyn's on the automation of teaching (Selwyn 2019) describe how analytics, scoring and prediction reconfigure institutional governance and learner subjectivity.

Algorithmic semiosis describes an adjacent regime: government by generation, in which the device produces the semiotic object itself, with its coherence and its self-evidence, upstream of any measurement. What datafication does to assessment, generation does to the object of study. The project offers itself as an extension of that field on the side of the production of forms, with the instruments semiotics has built for describing veridiction, inference and the interpretant. It connects in this respect to work carried out in Quebec on algorithmic cultures, notably Jonathan Roberge's on the social and cultural logics of machine learning (Roberge and Castelle 2021).

In **semiotics**, the project updates foundational concepts in the context of algorithmic production and opens onto a computational semiotics. In **media theory**, it connects the French tradition — Stiegler (Stiegler 2018), with algorithmic pre-tention as a deformation of tertiary retention; Simondon (Simondon 2024); Rancière — to the cognitive science of distributed cognition (Hutchins (Hutchins 1995), Norman (Norman 1991)). In **education research**, it proposes a shift from digital competences to semiotic formation, making resistance to algorithmic semiosis an academic competence in its own right.

The stance is descriptive and anti-hermeneutic, in Bertrand Gervais's sense (Gervais 2024): to describe the semiotic regimes that algorithmic outputs institute, and to reopen, through the second inquiry, the process their form claims to close. That description is the condition of any serious critical pedagogy.

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This summary describes work in progress; its formulations evolve with the research. For the long version, or to discuss the project, write to fdprudhomme@gmail.com.